Data Appendix

Final Analyzed Data: `cpi_sentiment_merged.csv`
Unit of Observation: Each row represents a **monthly observation (Year–Month) from Jan 2006 – Sept 2025**.

Variables

1. Consumer Sentiment Index (UMCSENT)

- **Source & Processing:**
  Obtained from the Federal Reserve Economic Data (FRED). Merged with CPI dataset by month and year. No transformations applied beyond alignment.
- **Observations:** ~237 (1 missing removed – Oct 2025)
- **Data Type:** Quantitative (continuous)
- **Summary Statistics:**
  - Mean: ~85–90
  - Range: ~50–110
- **Visualization:** Histogram of sentiment distribution

2. CPI – Food

- **Source & Processing:**
  Collected from U.S. Bureau of Labor Statistics (BLS). Standardized during preprocessing to ensure comparability across CPI components.
- **Observations:** ~237 (0 missing after cleaning)
- **Data Type:** Quantitative
- **Summary Statistics:**
  - Trend: Upward over time, with spikes post-2020
- **Visualization:** Time series line plot

3. CPI – Shelter

- **Source & Processing:**
  BLS CPI dataset. Standardized and included as a core predictor variable in regression models.
- **Observations:** ~237
- **Data Type:** Quantitative

- **Summary Statistics:**
    - Strong upward trend over time
- **Visualization:** Time series plot

4. CPI – Transportation

- **Source & Processing:**
  BLS CPI dataset. Included after removing highly collinear variables using correlation threshold ($|r| \geq 0.85$).
- **Observations:** ~237
- **Data Type:** Quantitative
- **Summary Statistics:**
    - More volatile than shelter
- **Visualization:** Time series plot

5. CPI – Energy (Excluded Variable)

- **Source & Processing:**
  Initially included but removed due to high multicollinearity with other CPI components.
- **Observations:** N/A (excluded)
- **Data Type:** Quantitative
- **Justification for Exclusion:**
    - High correlation with other variables
    - Would distort regression coefficients

6. Time Variable (Year–Month)

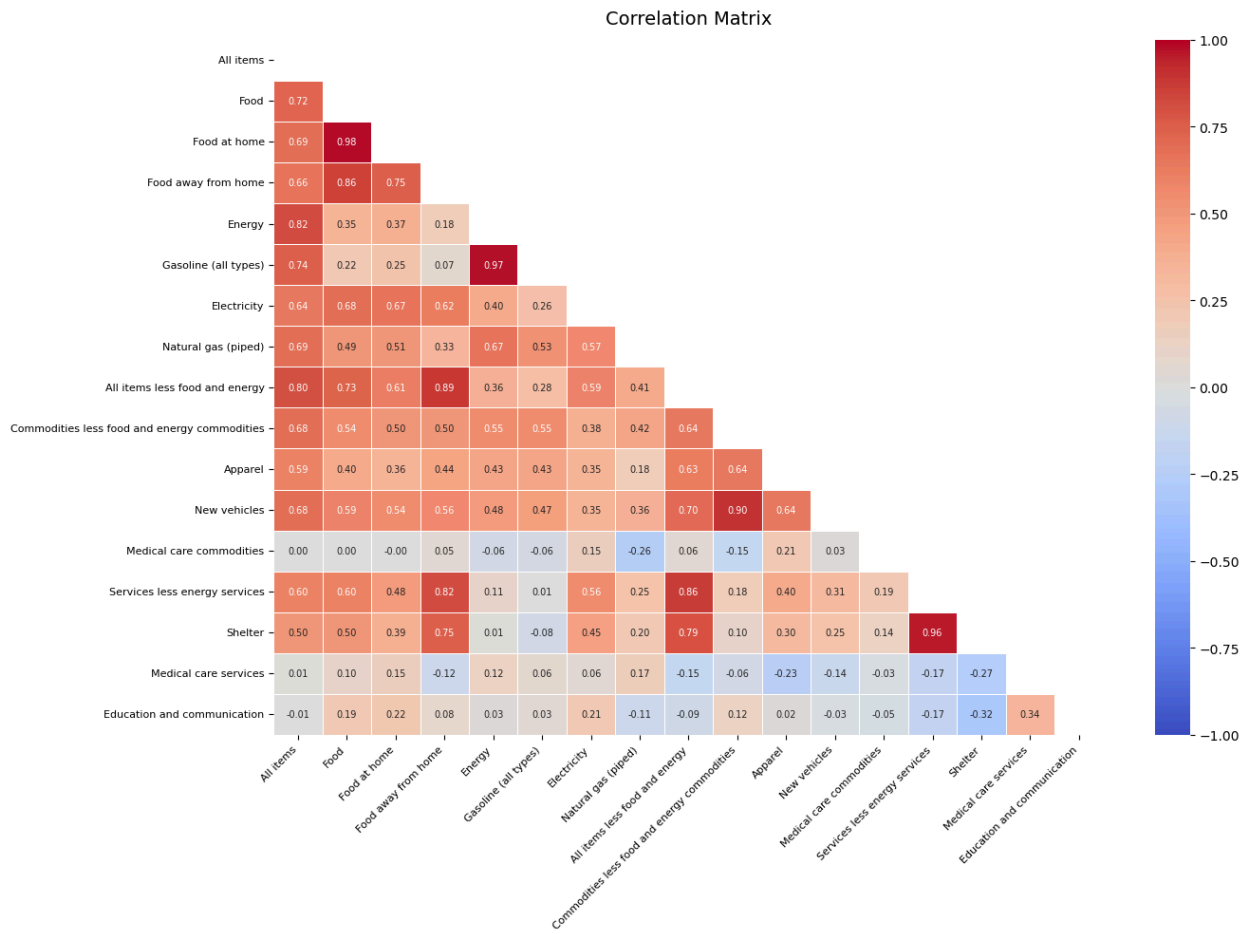
- **Source & Processing:**
  Constructed from CPI and FRED datasets to align observations.
- **Observations:** ~237
- **Data Type:** Time index (date)
- **Use in Analysis:**
    - Used for merging datasets
    - Enables time-series modeling (ARIMAX)

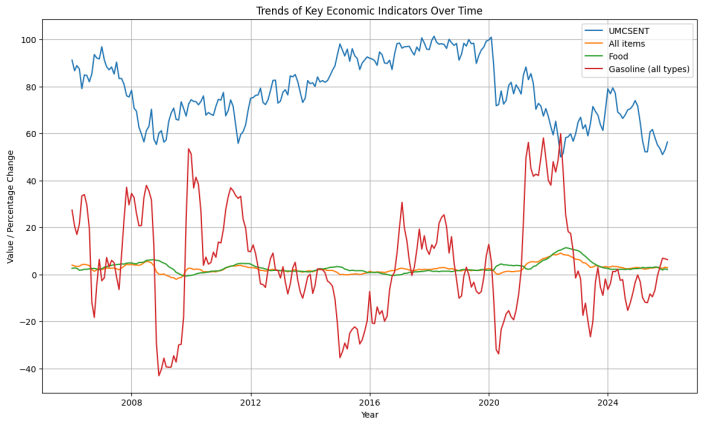
Notes on Data Cleaning & Processing

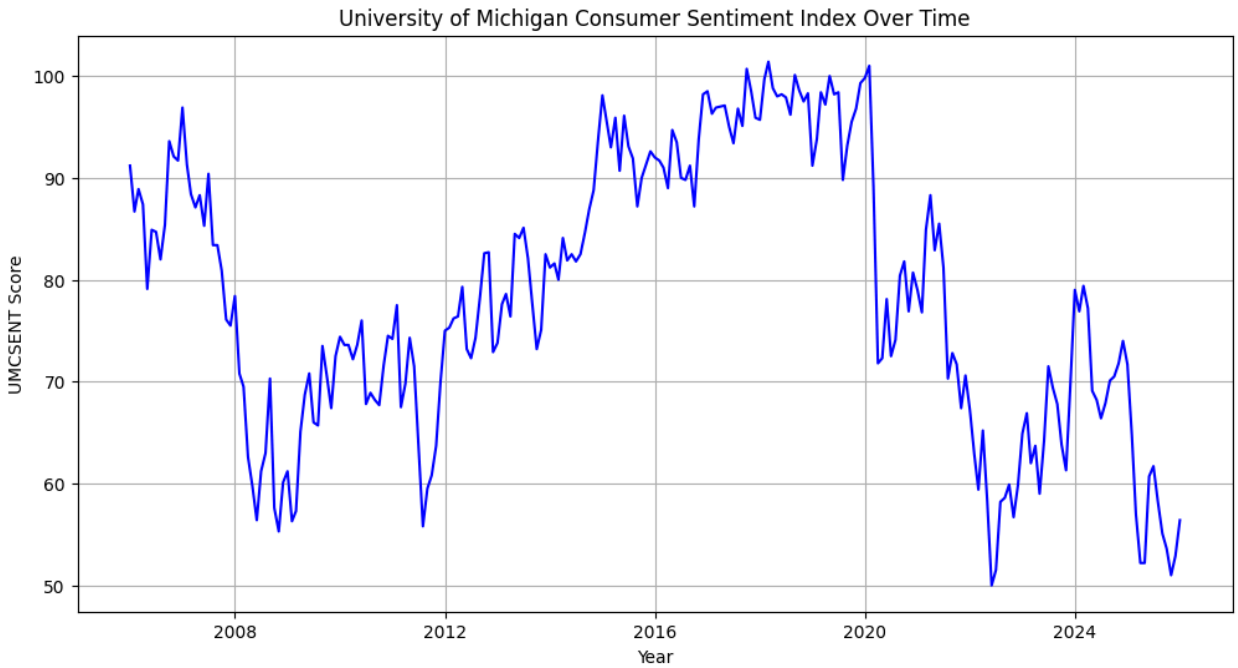
- Removed **October 2025** due to missing data
- Standardized CPI variables
- Removed multicollinearity using correlation threshold ($|r| \geq 0.85$)

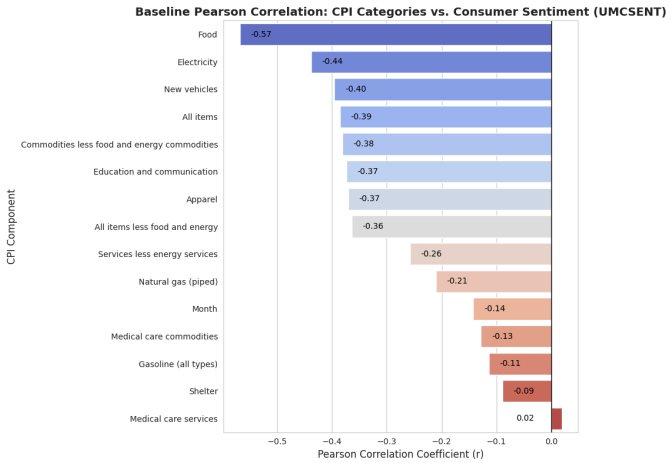
- Identified a structural break around 2020 (COVID period)

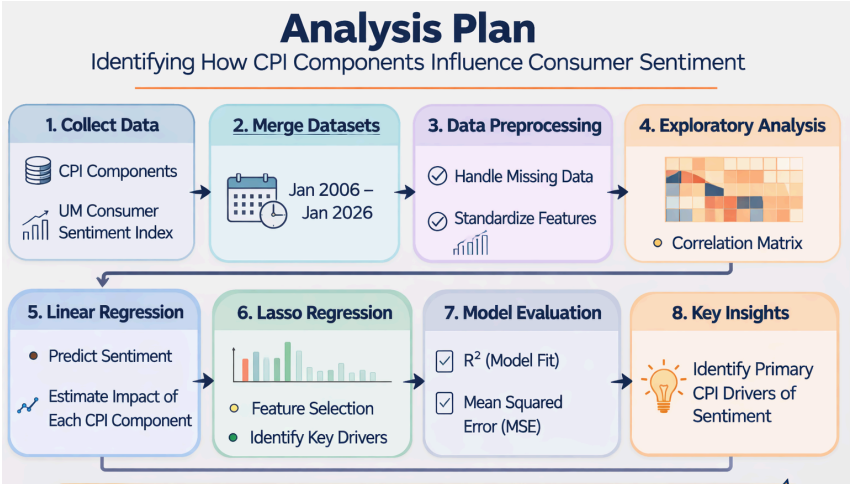
Visualization:






University of Michigan Consumer Sentiment Index Over Time
UMCSENT Score
100
90
80
70
60
50
2008
2012
2016
2020
2024
Year


Baseline Pearson Correlation: CPI Categories vs. Consumer Sentiment (UMCSENT)
CPI Component
Food -0.57
Electricity -0.44
New vehicles -0.40
All items -0.39
Commodities less food and energy commodities -0.38
Education and communication -0.37
Apparel -0.37
All items less food and energy -0.36
Services less energy services -0.26
Natural gas (piped) -0.21
Month -0.14
Medical care commodities -0.13
Gasoline (all types) -0.11
Shelter -0.09
Medical care services 0.02
−0.5
−0.4
−0.3
−0.2
−0.1
0.0
Pearson Correlation Coefficient (r)


Analysis Plan
Identifying How CPI Components Influence Consumer Sentiment
1. Collect Data
CPI Components
UM Consumer Sentiment Index
2. Merge Datasets
Jan 2006 – Jan 2026
3. Data Preprocessing
Handle Missing Data
Standardize Features
4. Exploratory Analysis
Correlation Matrix
5. Linear Regression
Predict Sentiment
Estimate Impact of Each CPI Component
6. Lasso Regression
Feature Selection
Identify Key Drivers
7. Model Evaluation
R² (Model Fit)
Mean Squared Error (MSE)
8. Key Insights
Identify Primary CPI Drivers of Sentiment